

JEPA 实验报告

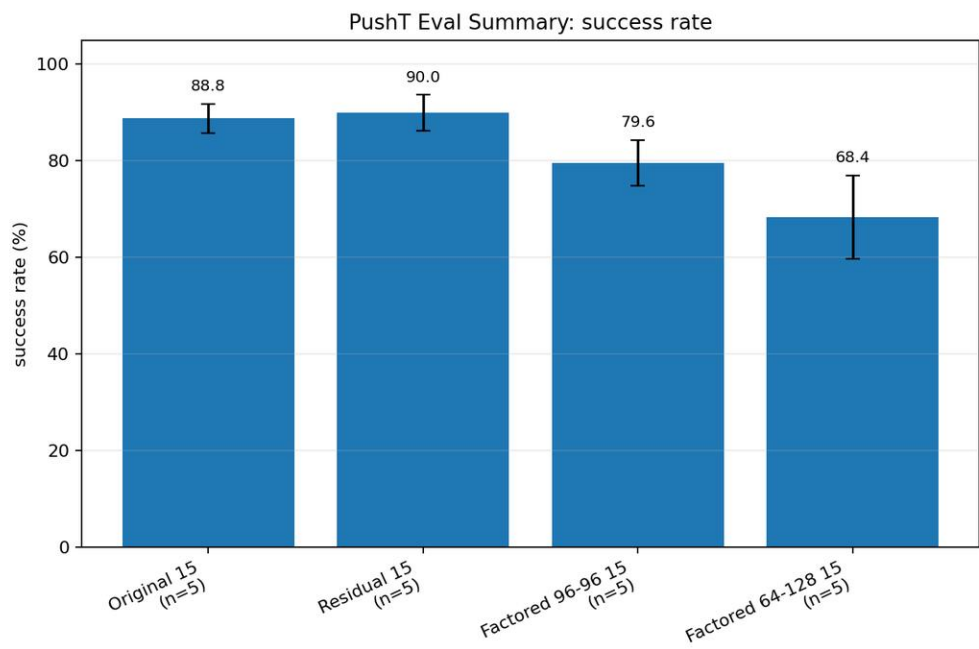
PushT / Reacher / Factored 结果汇总

本 PDF 基于 outputs/jepa_viz/experiment_summary_20260724 中的 eval 日志、训练 metrics.csv、latent 导出结果和 tools/jepa_viz 生成的可视化文件。

部分	内容
PushT	Original / Residual / Factored 独立对比
Reacher	Reacher Original / Residual 独立对比
Prediction	horizon cosine、alignment heatmap、condition ablation
Latent	active dims、pairwise cosine、covariance spectrum

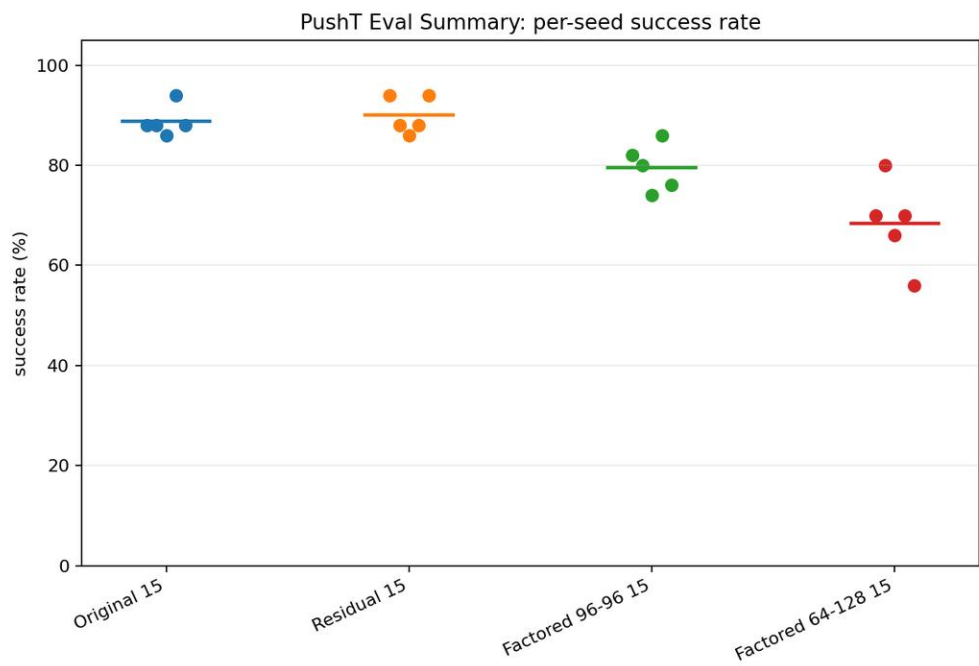
1. PushT : Original / Residual / Factored 对比

模型	成功率	Δ vs Original	time/ep
Original 15	88.80 +/- 3.03	+0.00	6.42s
Residual 15	90.00 +/- 3.74	+1.20	7.50s
Factored 96-96 15	79.60 +/- 4.77	-9.20	5.65s
Factored 64-128 15	68.40 +/- 8.65	-20.40	7.63s

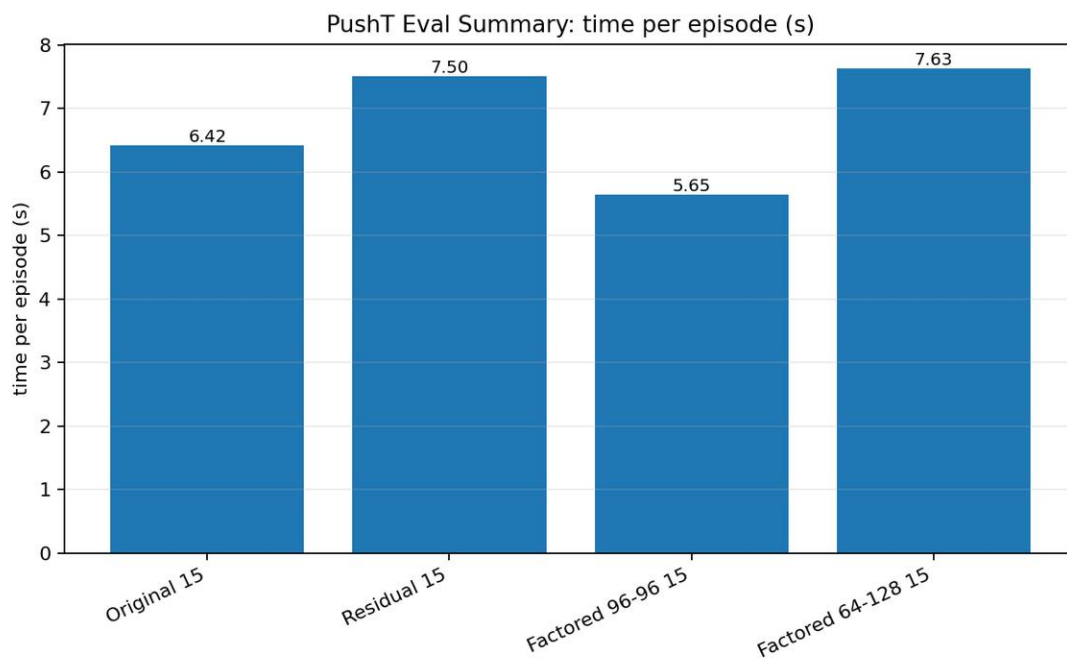


PushT success rate mean +/- std

指标	含义
success rate	多 seed 平均成功率，PushT 内横向比较。
Δ vs Original	相对 PushT Original 15 的成功率变化。
std	seed 间波动，越小越稳定。



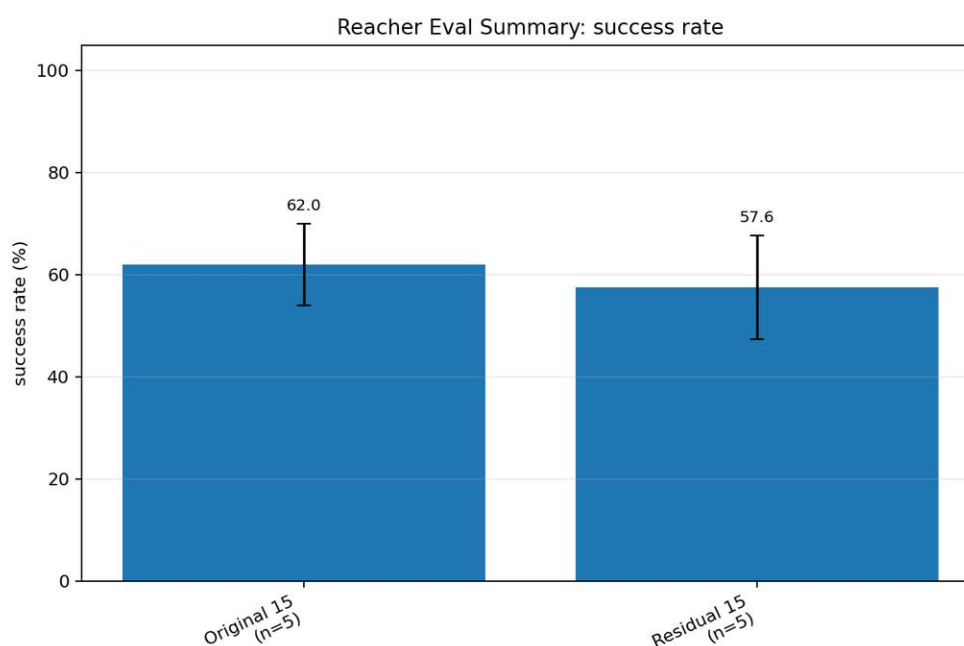
PushT per-seed success rate



PushT time per episode

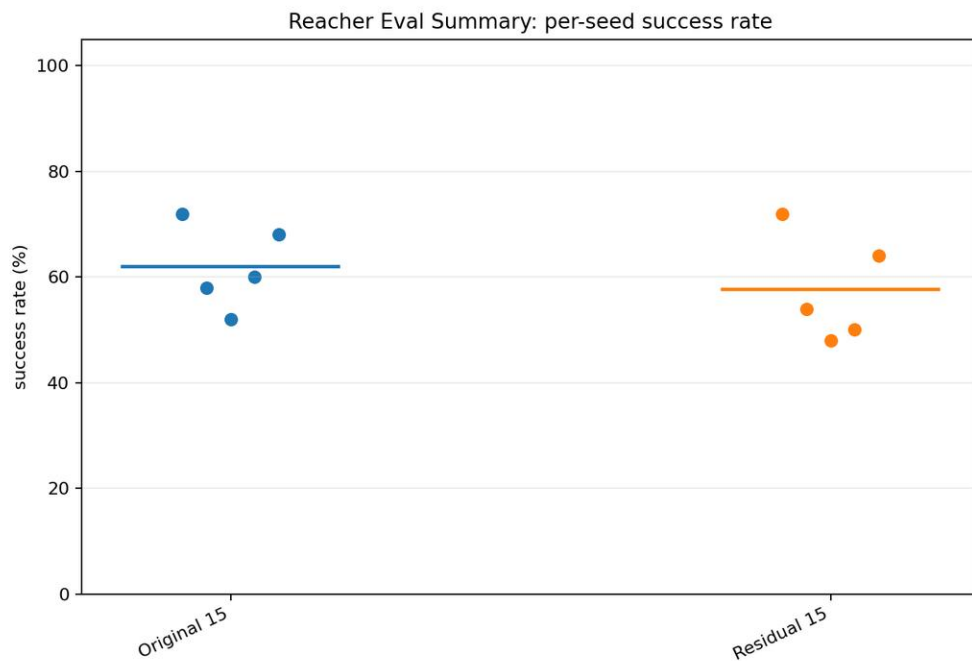
2. Reacher : Original / Residual 对比

模型	成功率	Δ vs Reacher Original	time/ep
Original 15	62.00 +/- 8.00	+0.00	4.16s
Residual 15	57.60 +/- 10.14	-4.40	4.12s

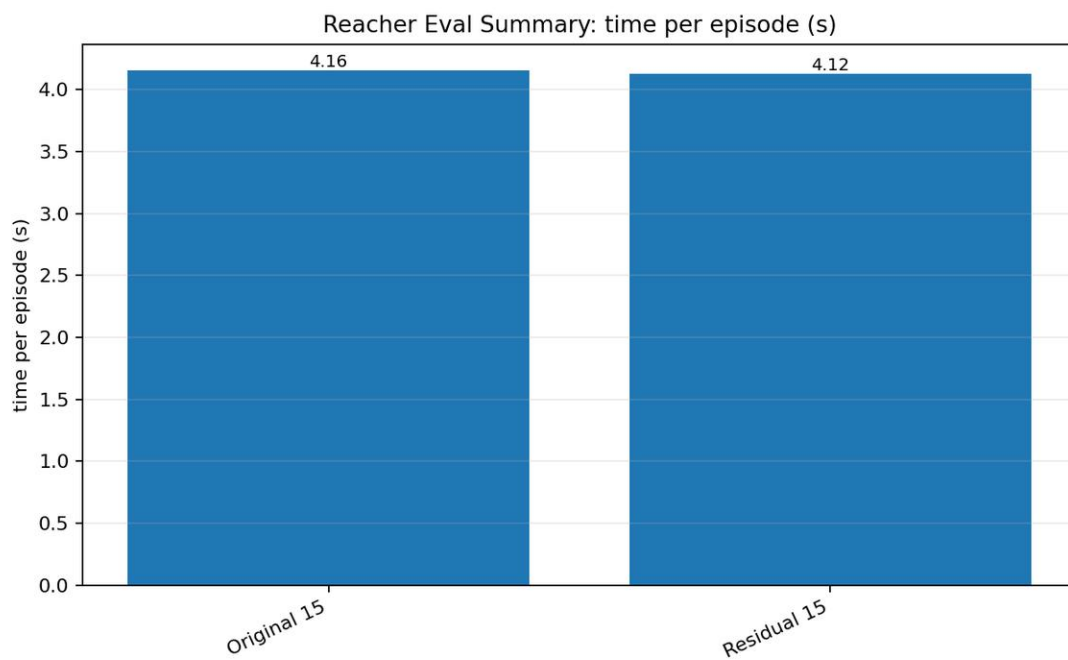


Reacher success rate mean +/- std

指标	含义
success rate	只在 Reacher Original 与 Reacher Residual 之间比较。
eval_budget	Reacher 使用 budget=50, 不和 PushT budget=300 耗时直接比较。
std	Reacher seed 间波动较大, 需要关注分布。



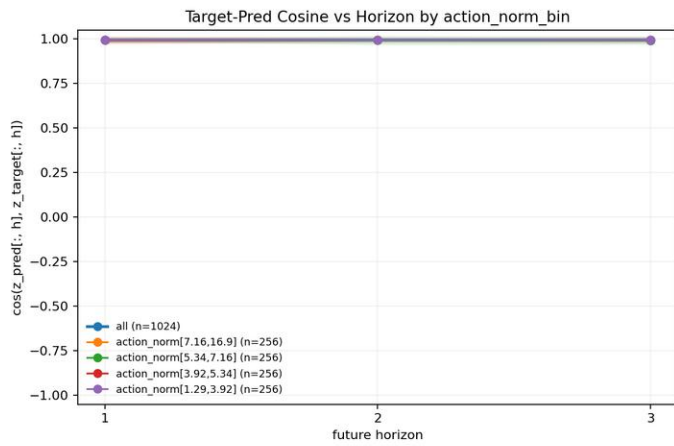
Reacher per-seed success rate



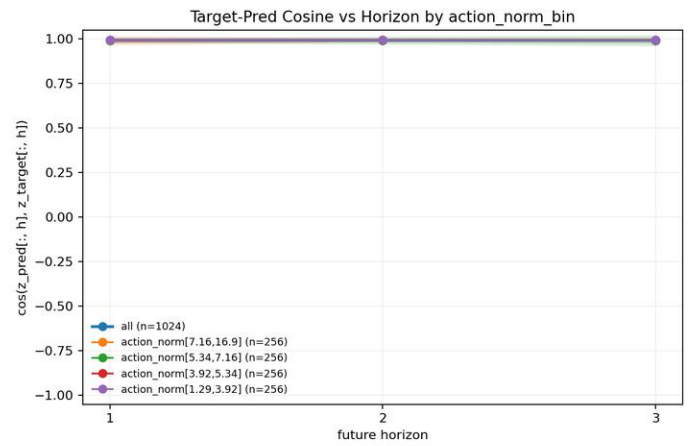
Reacher time per episode

3. PushT Prediction 指标

指标	Original	Residual
mean cosine	0.99399	0.99254
diagonal gap	0.20243	0.16525
top-1 horizon	98.70%	96.61%
normal MSE	0.01070	0.01311
shuffled-action MSE	0.37885	0.28743

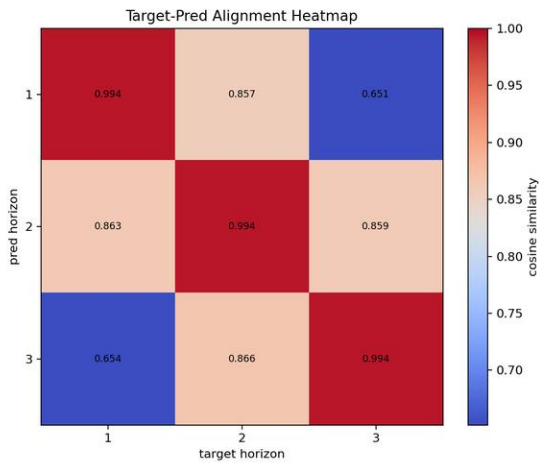


Original horizon cosine

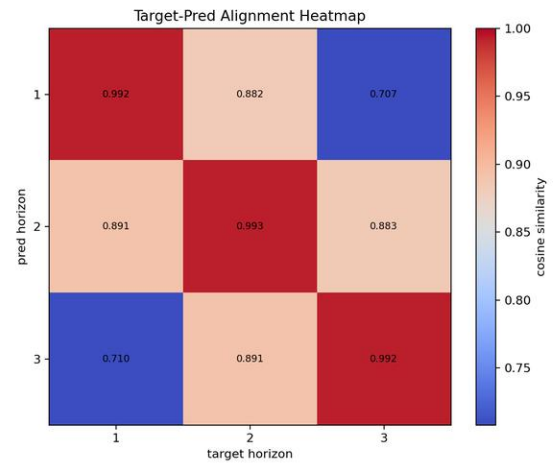


Residual horizon cosine

指标	含义
horizon cosine	$\cos(z_{\text{pred}}, z_{\text{target}})$, 越高表示预测越接近目标。
action_norm_bin	按 action 范数分桶, 观察动作强度对预测的影响。

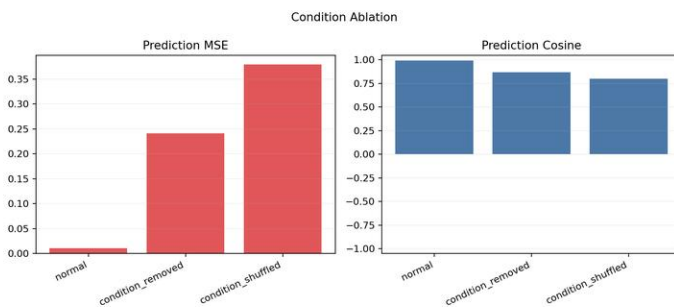


Original alignment heatmap

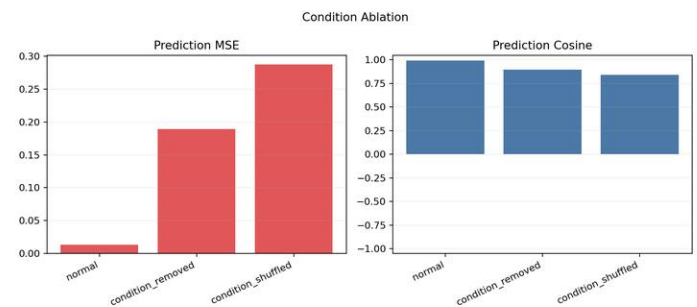


Residual alignment heatmap

指标	含义
diagonal gap	对角线平均值减去非对角线平均值, 越大越能区分 future horizon。
top-1 matching	预测 horizon 匹配正确 target horizon 的比例。



Original condition ablation

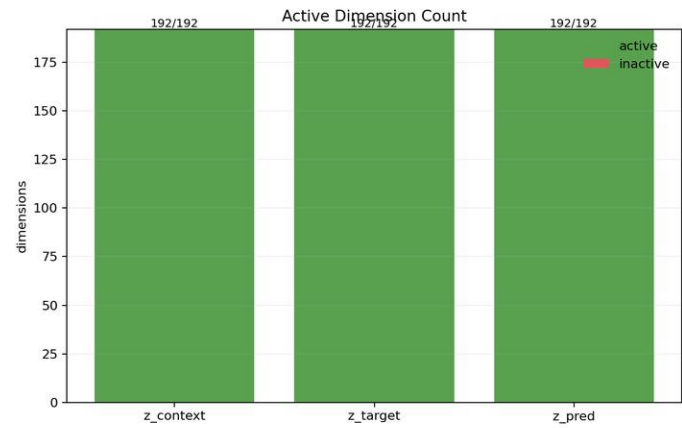


Residual condition ablation

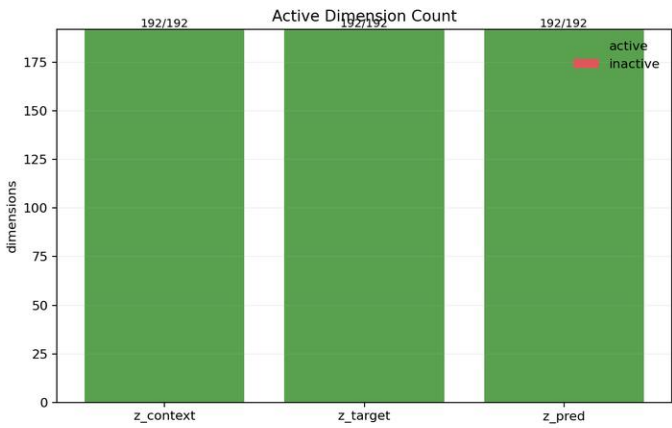
指标	含义
normal	使用正常 action condition 的预测。
condition shuffled	打乱 action 后的预测; 退化越明显, 说明模型越依赖正确 action。

4. PushT Latent 表征诊断

指标	Original z_pred	Residual z_pred
active dims	192/192	192/192
effective rank	59.59	58.32
participation ratio	54.41	52.93
top10 explained var	0.257	0.266
pairwise cosine q95	0.2357	0.2302

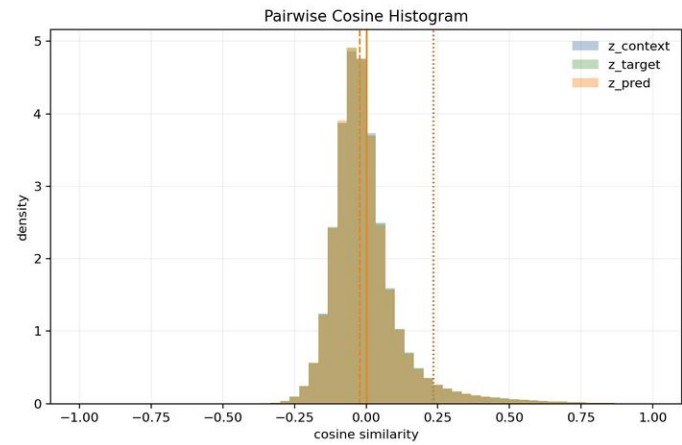


Original active dims

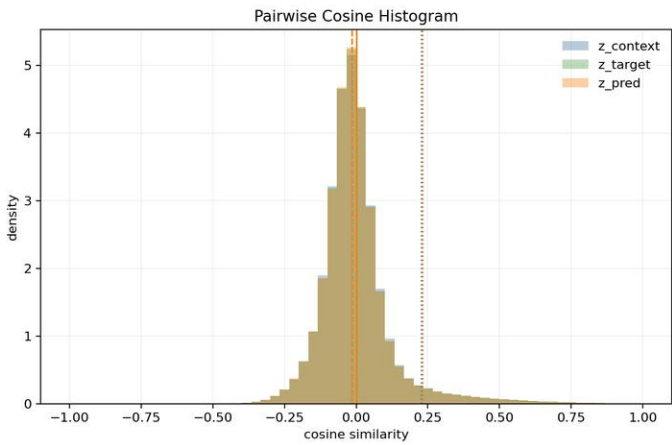


Residual active dims

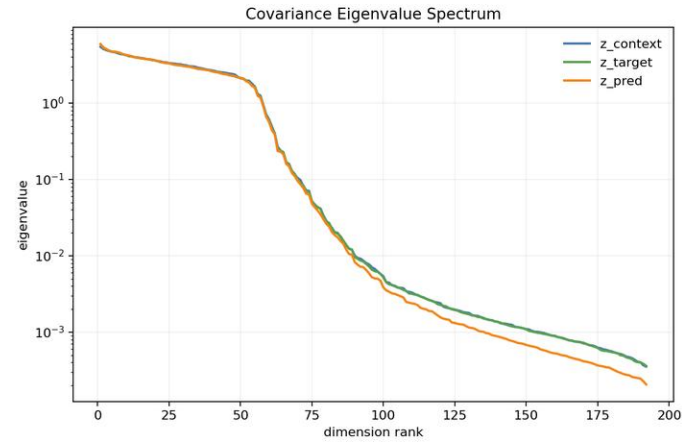
指标	含义
active dims	std 超过阈值的 latent 维度数；过低表示 collapse。
effective rank	协方差谱有效秩，越高表示维度利用更分散。



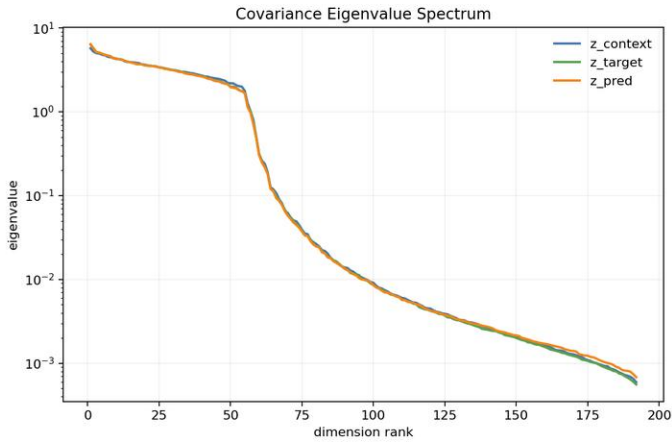
Original pairwise cosine



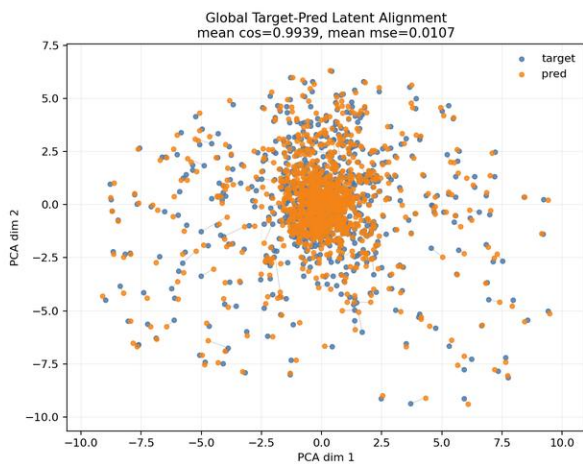
Residual pairwise cosine



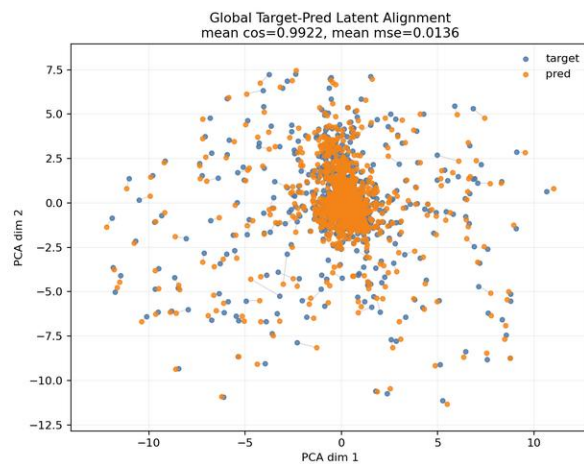
Original covariance spectrum



Residual covariance spectrum



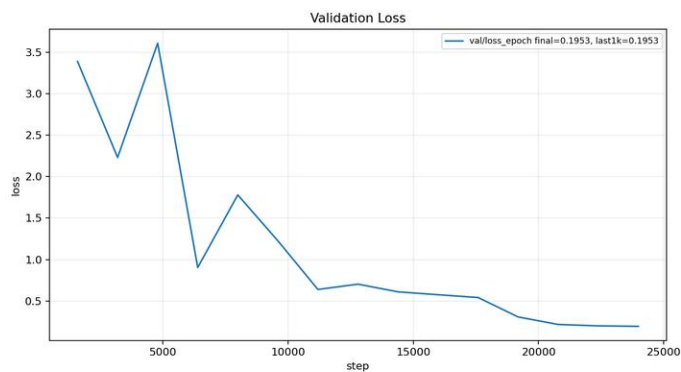
Original target-pred alignment



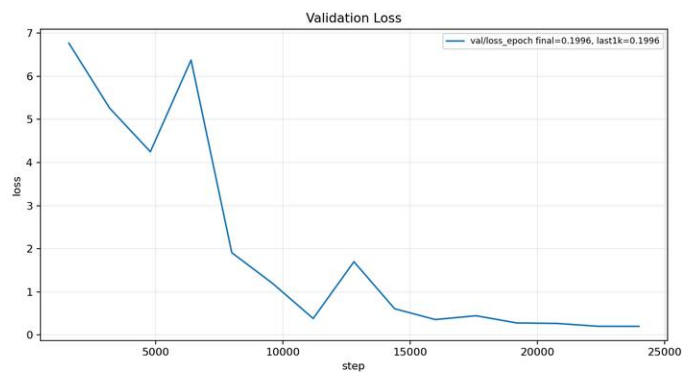
Residual target-pred alignment

5. 训练曲线索引

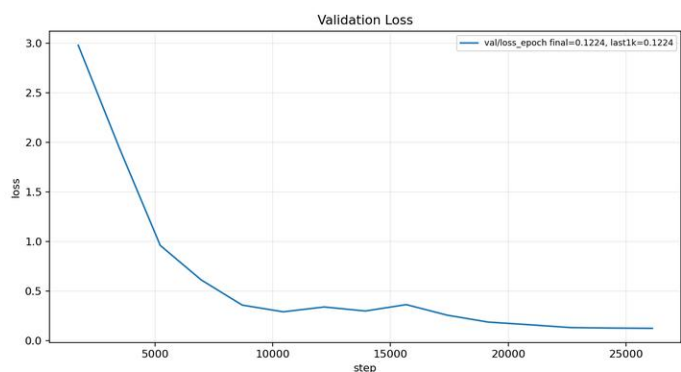
模型	主要曲线目录
Reacher Original 15	training/reacher_original_15/
Reacher Residual 15	training/reacher_residual_15/
Factored 96/96 15	training/factored_96_96_15/
Factored 64/128 15	training/factored_64_128_15/



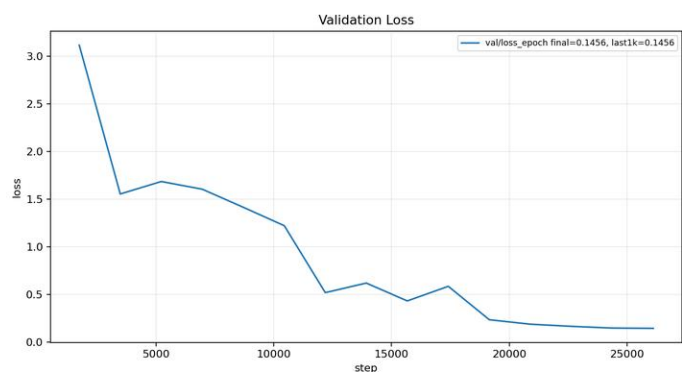
Reacher Original val loss



Reacher Residual val loss



Factored 96/96 val loss



Factored 64/128 val loss

指标	含义
val loss	验证集 loss，用于观察泛化。
per-horizon MSE	每个 future horizon 的 MSE，检查多步预测是否区分。
samples_per_sec	训练吞吐，不与 learning rate 混在同一 y 轴。

6. 结果说明

PushT 上 Residual 15 的成功率均值为 90.0%，Original 15 为 88.8%，差距为 +1.2 个百分点。Factored 两个版本均低于 PushT Original/Residual，其中 96/96 明显优于 64/128。

Reacher 是独立实验，只比较 Reacher Original 与 Reacher Residual。当前 Reacher Residual 15 为 57.6%，Reacher Original 15 为 62.0%，Residual 未带来改善。

PushT latent prediction 中，Original 的 normal MSE 更低、diagonal gap 更大，说明 horizon alignment 更清晰。两个 prediction report 都提示 mean cosine > 0.99，虽然 z_pred 与 z_target 不是完全相同，但后续仍应持续检查 target leakage 或 latent 读取路径。